ECHO-7

Visual bay finding and docking | Product & operating guide

DE1-SoC / OV7670 / UGV02

Choose a blue bay marker and start a task. ECHO-7 searches, confirms the marker, aligns, approaches and parks, with live VGA feedback, remote stop and front sonar protection.

OV7670 camera — Live colour image → **DE1-SoC FPGA** — Cyclone V 5CSEMA5F31C6 | vision + control → **UGV02 chassis** — Differential-wheel motion

Also fitted: J3 RCWL-1601 sonar, NEC IR receiver/remote, VGA monitor, switches, keys, LEDs and six-digit display. USB-Blaster provides PC download/debug. Chassis UART: 115200 8N1, 3.3 V logic, common ground.

THREE OPERATING MODES

MODE / DISPLAY CODE	PURPOSE
0 - INTEGRATED	Camera-guided driving. Choose one bay; POWER or red2 starts a new task. Default after power-up/reset, with motion disarmed.
1 - VISION_ONLY	Live recognition, tracking, calibration and VGA; wheel commands stay zero. Use for marker and camera setup.
2 - CHASSIS_INJECT	PC/JTAG virtual targets test the same controller. Real IR and sonar stay active; loss of the PC injection session stops motion.

Choose modes with the JTAG debug tool. Changing mode stops the task. INTEGRATED needs no PC heartbeat.

RED AND BLUE MARKERS

Blue 1-6: destination bays. SW0-SW5 choose one number to follow. Keep its complete border and white anchor visible.

Red1: colour calibration. Hold red1 and its paired blue reference stripe steady. Motion pauses until calibration finishes: **CAL:Y**. The stripe label **BJ** is not a bay.

Red2: reselect and start. Re-read switches, relock and request a task. Remove it, then present it beside the target. It cannot choose a bay or restart an active task. Keep hands/cards clear of J3.

SWITCHES AND BUTTONS

SW0-SW5	SW0/1/2/3/4/5 select blue 1/2/3/4/5/6. **Exactly one ON** to start. Changing selection cancels motion.
SW6	Candidate colour/number diagnostics.
SW7	Centre colour sample and marker identity.
SW8	0: normal perspective handling. 1: bounding-box recognition bypass.
SW9	Camera-only display; cancels motion. OFF requires a new start to move again.
KEY0 / KEY1	KEY0: full reset. KEY1: next display page.

Remote POWER: start. **A:** latched emergency stop. After A, one fresh POWER releases it; release and press again to start. Holding POWER cannot do both.

PC/JTAG: In stopped, disarmed VISION_ONLY, use **tune** for speeds/centring and **pulse** for short turns. Ranges are not validated presets. Commands: README.

WHAT THE VGA IMAGE SHOWS

Upright camera image with red/blue contours and labels **R1/R2**, **B1-B6** or **BJ**. **CAL:W/B/Y** = waiting/calibrating/calibrated. **FRAME:GOOD/BAD/STALE** = image health. **STATE** = calibration interaction, not navigation.

SW6: RGB/HSV, colour, number or UNKNOWN, rejection code E. PAGE:1/2 alternates every 2 s across 16 slots, identified by yellow sample marks.

SW7: white 8 x 8 centre sample: RGB/HSV and the complete marker covering it. **SW8:** BYPASS:1 changes recognition, not camera view or safety. **SW9:** hides overlays.

READING THE SIX-DIGIT DISPLAY

page | mode | four data digits

d1 0520 = distance page, VISION_ONLY, **520 mm**. The second digit is the mode code above, not the bay number.

d	Millimetres. **FAr** = valid out-of-range, not measured 4 m. Four blank digits = invalid distance.
H	Last IR word, upper half in hexadecimal. Right pair **12** = POWER, **0F** = A (examples ED12 / F00F).
L	Last IR word, lower/address half in hexadecimal. H/L are blank until a complete valid IR word arrives.

KEY1: d - H - L - d. Left digit **n** = stable no-selection; **C** = stable multi-switch conflict, replacing the page letter. No ARM or driving-state page.

LEDs: 0 motion permitted; 1 IR stop latched; 3 docked; 8 target locked and fresh.

APPROACH / STOP Defaults: 5 cm/s; 3 cm/s within 500 mm. A lock within +/-30 centre pixels allows straight approach; dock at <=350 mm, then require a new start. Stop for <=250 mm, invalid/stale range or IR A. HOLD releases after three fresh >=300 mm readings. After target loss, stop for 1.5 s and reconfirm fresh frames.

ADJUSTABLE SHORT TURN ALIGN only: kick at 3-8 cm/s for 0-100 ms, then cruise at 3-5 cm/s (3 cm/s for small errors). Set total min/max to 20-300 ms (min <= max); time settings use 10 ms steps. Total includes/truncates kick; 0 ms disables it. Centring, loss or safety can end it early. Defaults: kick off (5 cm/s), total 20-100 ms. SEARCH/APPROACH have no kick.

SEARCH / RESET Search is not guaranteed 360 degrees. Stop to confirm sightings; loss can trigger up to three 150 ms reverse steps, each reconfirming within the scan budget. Recalibrate after changing search speed. Reset/power loss restores defaults and clears colour/scan calibration. Restore scan calibration for unseen-target search; visible locks can still approach.